AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Flexential Minneapolis - Chaska, MN -- station KOEO, America/Chicago
Building	OSM 52227492 Flexential Minneapolis - Chaska
Facade gap	not applicable -- one building, no second facade
Weather record	42,558 real hourly records from KOEO
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-05-04 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 9 of 23 hours with 1 mode change. 9 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	9 of 23 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	9 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s)

9 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
01	mechanical	yes	switch budget	9.390	18.0	9.110	4.260
02	mechanical	yes	switch budget	9.610	18.0	8.110	3.658
03	mechanical	yes	switch budget	9.830	18.0	8.000	3.300
04	mechanical	yes	switch budget	10.110	18.0	8.110	3.243
05	mechanical	yes	switch budget	10.220	18.0	6.390	3.086

06	mechanical	yes	switch budget	10.610	18.0	5.560	2.865
07	mechanical	yes	switch budget	12.330	18.0	7.220	3.469
08	mechanical	yes	switch budget	14.000	18.0	8.330	4.361
09	mechanical	yes	switch budget	16.780	18.0	8.780	5.467
10	mechanical	no	dry-bulb	19.050	18.0	10.280	6.036
11	mechanical	no	dry-bulb	19.220	18.0	11.890	6.533
12	mechanical	no	dry-bulb	19.170	18.0	12.110	6.611
13	mechanical	no	dry-bulb	19.780	18.0	12.610	5.714
14	mechanical	no	dry-bulb	18.830	18.0	12.890	4.754
15	FREE	yes	-	16.890	18.0	13.110	3.849
16	FREE	yes	-	16.170	18.0	13.330	3.262
17	FREE	yes	-	15.830	18.0	14.000	3.095
18	FREE	yes	-	14.160	18.0	13.330	3.063
19	FREE	yes	-	13.000	18.0	11.780	3.174
20	FREE	yes	-	12.450	18.0	9.890	3.809
21	FREE	yes	-	12.500	18.0	9.110	4.555
22	FREE	yes	-	12.720	18.0	8.220	4.863
23	FREE	yes	-	13.780	18.0	3.720	5.100

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.610 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.830 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.610 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.050 C against a 18.0 C limit, so it fails by 1.050 C. It would take a limit 1.050 C higher, or a bound 1.050 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.220 C against a 18.0 C limit, so it fails by 1.220 C. It would take a limit 1.220 C higher, or a bound 1.220 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.170 C against a 18.0 C limit, so it fails by 1.170 C. It would take a limit 1.170 C higher, or a bound 1.170 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.780 C against a 18.0 C limit, so it fails by 1.780 C. It would take a limit 1.780 C higher, or a bound 1.780 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.830 C against a 18.0 C limit, so it fails by 0.830 C. It would take a limit 0.830 C higher, or a bound 0.830 C tighter, to change this hour.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.890 C, which is 1.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.849 C, and the margin is measured, not chosen: 3.849 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.170 C, which is 1.830 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.262 C, and the margin is measured, not chosen: 3.262 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.830 C, which is 2.170 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.095 C, and the margin is measured, not chosen: 3.095 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.160 C, which is 3.840 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.063 C, and the margin is measured, not chosen: 3.063 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.000 C, which is 5.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.174 C, and the margin is measured, not chosen: 3.174 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.450 C, which is 5.550 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.809 C, and the margin is measured, not chosen: 3.809 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.500 C, which is 5.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.555 C, and the margin is measured, not chosen: 4.555 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.720 C, which is 5.280 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.863 C, and the margin is measured, not chosen: 4.863 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.780 C, which is 4.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.100 C, and the margin is measured, not chosen: 5.100 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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